Modulus Functions

Questions 1–56

Question 1. For the functions $f(x) = |x - 2|$ and $g(x) = |x| - 2$, find

a) $f(1)$
b) $f(-3)$
c) $f(0)$
d) $g(1)$
e) $g(0)$
f) $g(-4)$

Question 2. If $|x| = 4$ find the possible values of $|2x - 5|$.

Question 3. If $|x| = 2$ find the possible values of $|4x + 1|$.

Question 4. Write down the y-intercept of each of the graphs of the following functions:

a) $y = |x + 2|$
b) $y = |x - 1|$
c) $y = |x - 3| + 5$
d) $y = 2|x + 4| + 8$
e) $y = -5|x - 2| + 1$
f) $y = 6 - 4|x + \frac{1}{2}|$

Question 5. Write down the vertex of the graph of each of the following functions:

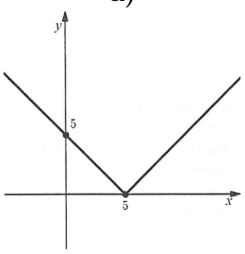
a) $y = |x + 3|$
b) $y = |x - 2|$
c) $y = |x - 1| + 5$
d) $y = |x + 3| - 8$
e) $y = 2|x + 4| + 13$
f) $y = -8|x - 1| + 9$
g) $y = 2 - 7|x + \frac{1}{2}|$

Question 6. Sketch each of the following, labelling only the vertex and the y-intercept:

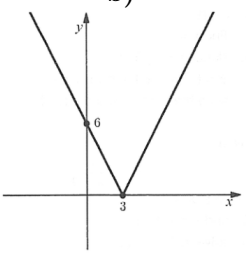
a) $y = |x + 3|$
b) $y = |x - 2| + 4$
c) $y = |x + 1| - 1$
d) $y = |x - 7| - 7$
e) $y = 2|x + 1| + 3$
f) $y = 3|x - 5| + 1$
g) $y = 2|x + 9|$
h) $y = -4|x - 1|$
i) $y = 7 - |x - 6|$
j) $y = 9 - 2|x + 4|$

Question 7. For each of the following graphs, state the equations:

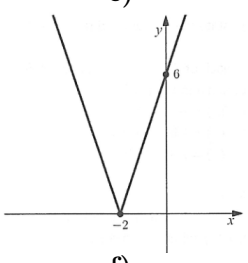
a)



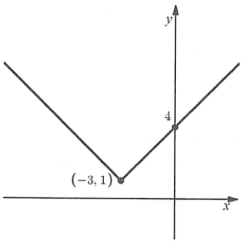
b)



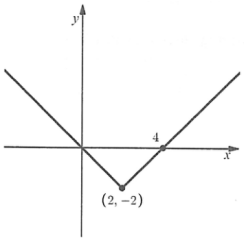
c)



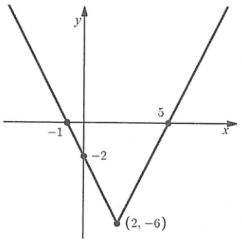
d)



e)

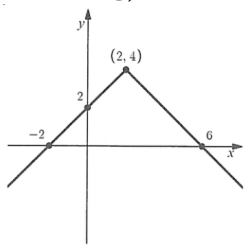


f)

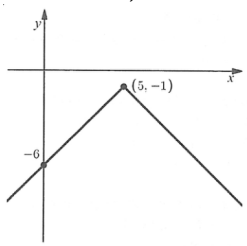


Question 7 (continued). For each of the following graphs, state the equations:

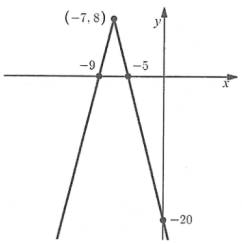
g)



h)



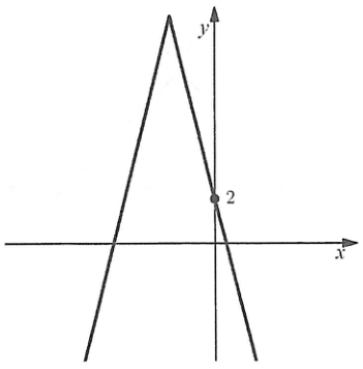
i)



Question 8. Solve each of the following equations:

a) $|x+3|=5$
b) $|x-2|=7$
c) $|2x+1|=4$
d) $-|x+5|=-3$
e) $|x-5|=-3$

Question 9. The diagram shows a sketch of $f(x)=k-4|x+2|$



Find

a) the value of k
b) the values where $f(x)$ intersects with the x-axis.

Question 10. Solve each of the following equations:

a) $|x-1| = |x+8|$
b) $|x+3| = |x-4|$
c) $|x-10| = |x-1|$

Question 11. Solve each of the following equations:

a) $|x+5| = 2|x+3|$
b) $|x+6| = 3|x-3|$
c) $|x+2| = 5|x+8|$
d) $|x+2| = 10 - |x-5|$
e) $|x-4| - 8 = -|x-3|$
f) $|x+4| = 4 - |x-3|$

Question 12. Write the following as a modulus inequality:

a) $-6 < x < 6$

b) $-4 < x < 6$

c) $-3 < x < 25$

d) $x < 1, x > 5$

Question 13. Write $6 \leq x \leq 14$ in the form $|2x - a| \leq b$ where a and b are integers to be found.

Question 14. Sketch the graph of $y = 6 - |2x - 7|$ and $y = 3$ on the same axes, and hence solve the inequality $6 - |2x - 7| > 3$.

Question 15. Sketch $y = |2x|$ and $y = |3x - 9|$ on the same axes and hence solve $|2x| \geq |3x - 9|$.

Question 16. Find the sets of values of x for which, giving your answer in interval notation:

a) $|x+3| > 10$

b) $4 - |x| \leq |x-2|$

c) $|4x| - 2 \geq |7-2x|$

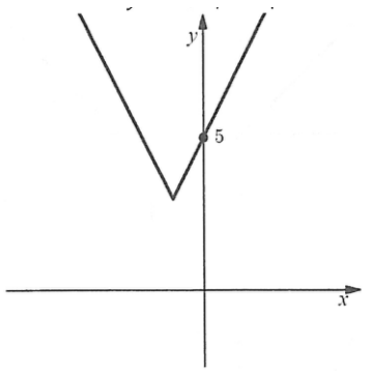
Question 17. Sketch the graph of $y = |3x+k|$ where k is a positive constant. Include all points of intersection with the coordinate axes.

Question 18. A function is given to be $f(x) = \frac{7}{8}|x+1| + 4$

a) Sketch $f(x)$

b) Solve $f(x) = 2x + 3$

Question 19. The diagram shows a sketch of $y = k + 2|x+1|$



a) Find the value of k

b) For value of k solve $k + 2|x+1| = x + 8$

Question 20. Find the values of x such that

a) $|x| + 4 \geq |x + 4|$

b) $-2|x| + 8 > -2|x + 4|$

Question 21. For the following, find the possible values of k for which there are two distinct solutions:

a) $2|x - 1| + 6 = k$

b) $3|4 - x| - 4 = k$

c) $8 - 2|x + 1| = k$

Question 22. It is given that $f(x) = |x + 4|$

a) Sketch $f(x)$

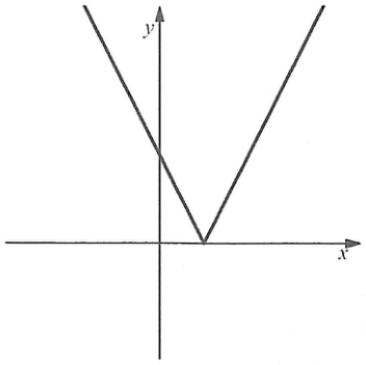
b) Find the values of k, such that $f(x) = kx$ has 2 distinct solutions.

Question 23. It is given that $f(x) = 4 + |x - 2|$

a) Sketch $f(x)$

b) Find the values of k such that $f(x) = 4 + kx$ has 2 distinct solutions.

Question 24. The diagram shows a sketch of $y = |2x - 5|$



a) State the coordinates of the points of intersection with the axes.
b) The graph of $y = kx + 2$ has two distinct points of intersection, find the possible values of k
c) State the solutions of $|2x - 5| = kx + 2$ giving your answer in terms of k

Question 25. Describe the transformations that are required to transform the graph of $y = |x + 3|$ onto the graph of

a) $y = 2|x + 3|$
b) $y = |x - 2|$
c) $y = |x + 5|$
d) $y = |2x + 6|$
e) $y = |2x + 3|$

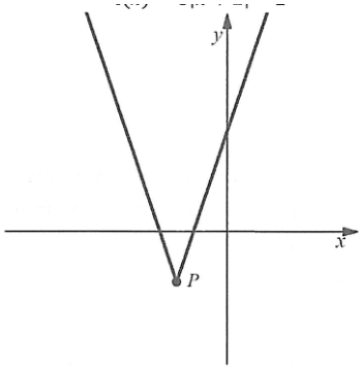
Question 26. Describe the transformations that are required to transform the graph of $y = |2x - 1|$ onto the graph of

a) $y = 3|2x - 1|$
b) $y = |6x - 3|$
c) $y = |2x - 3|$
d) $y = |2x + 5| + 4$
e) $y = |2x - 4| - 1$

Question 27. Describe the transformations that are required to transform the graph of $y = |ax + b|$ onto the graph of

a) $y = \frac{1}{2}|ax + b|$
b) $y = |\frac{1}{3}ax + b|$
c) $y = |ax + b + 4a| - 5$

Question 28. The diagram shows a sketch of $f(x) = 3|x + 2| - 2$



The point P is at the vertex of $f(x)$

a) State the coordinates of P

b) Solve the equation $3|x + 2| - 2 = -3x + 18$

A line $y = kx$ where k is a constant.

c) Find the possible values of k such that it intersects $y = 3|x + 2| - 2$ at least once.

Question 29. A function is given to be $f(x) = k - 3|x + 2|$ where $k \geq 6$

a) Sketch $f(x)$ including labels for all the points of intersection

b) Find the solutions to $k - 3|x + 2| = x + 4$ giving your answer in terms of k

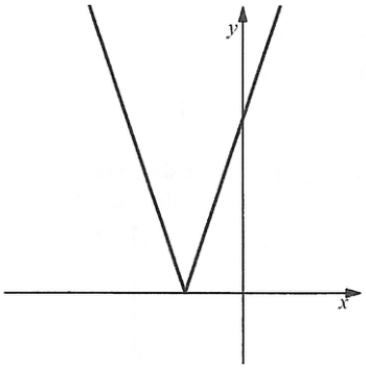
Question 30.

a) Sketch on a single diagram $y = |x + 2|$ and $y = \frac{k}{x}$ where k is a positive constant.

b) Explain why $|x + 2| = \frac{k}{x}$ has exactly one solution for any positive value of k

c) Find the solution to $x|x + 2| = k$ where $k > 0$, giving your answer in terms of k

Question 31. The diagram shows a sketch of $f(x) = 3|x + 2k|$ where $k \geq 0$



Sketch the graph of each of the following, including the coordinate of the vertex and the y-intercept in each case

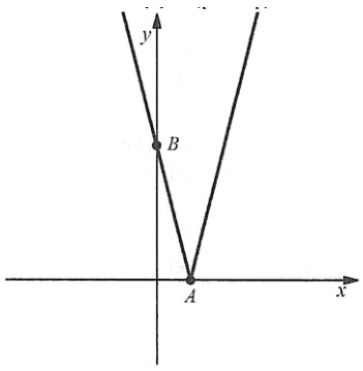
a) $y = -3|x + 2k|$

b) $y = 3|x + 2k| + 4$

Question 32. Given that $f(x) = k + |x - 2|$ where $k \geq 0$, sketch each of the following, labelling any intersections with the axes:

a) $y = f(x)$
b) $y = -f(x)$
c) $y = 2f(x)$

Question 33. The diagram shows a sketch of $f(x) = |px + q|$



The point A is at the minimum of the graph and lies on the x-axis. The point B is where the graph intersects with the y-axis. The line $y = 4x + k$ is parallel to part of the graph and $|px + q| = 4x + k$ has no solutions for $k < -9$. Find the values of p and q.

Question 34. State the number of solutions to the equation $|x+2|+|x|=1$.

Question 35. State the range each of the following, giving your answer in set notation:

a) $f(x)=|x+1|$
b) $f(x)=-|x+3|$
c) $f(x)=|x-1|+3$
d) $f(x)=2|x+4|-1$
e) $f(x)=5-|x-3|$
f) $f(x)=1-2|x+7|$

Question 36. It is given that $f(x) = 10|x - 2| - 12$ where $x \geq 0$

a) Sketch $f(x)$

b) State the range of $f(x)$ and domain of $f(x)$

c) Find the values of x such that $f(x) \leq -\frac{1}{2}x + \frac{13}{4}$

Question 37. It is given that $f(x) = 3|5 - x| + 5$, $x \geq 0$

a) Sketch $f(x)$

b) State the range of $f(x)$

c) Solve $f(x) = \frac{1}{3}x + 15$

Question 38. For $f(x) = 2x + 1$, $g(x) = |x|$ and $h(x) = -|x|$, find:

a) $fg(x)$
b) $gf(x)$
c) $hf(x)$
d) $hfg(x)$

Question 39. For $f(x) = x^2 + 1$, $g(x) = |x|$ and $h(x) = -|x|$, find:

a) $fg(x)$
b) $gf(x)$
c) $hf(x)$
d) $fh(x)$
e) $hg(x)$

Question 40.

a) Sketch on a single diagram $y = |x + 1|$ and $y = k \ln x$ where k is a negative constant.

b) Explain why $|x + 1| = k \ln x$ has exactly one solution for any negative value of k

c) Starting with $x_0 = 1$, use the Newton-Raphson method to find the solution to $|x + 1| = -5 \ln x$ giving your answer to 3 significant figures.

Question 41.

a) Sketch on a single diagram $y = 2|x + 1|$ and $y = ke^x$ where k is a positive constant and $k > 2$

b) Explain why $2|x + 1| = ke^x$ has exactly one solution

c) Starting with $x_0 = -2$, use the Newton-Raphson method to find the solution to $|x + 1| = 2e^x$ giving your answer to 3 significant figures.

Question 42. Sketch each of the following:

a) $y = 2|x| - 5$
b) $y = 4|x|^2 - 5$
c) $y = 3|x|^2 - 4|x|$
d) $y = 2|x|^2 - 3|x| - 5$
e) $y = -|x|^2 + 4|x| + 5$
f) $y = \ln|x|$
g) $y = \frac{1}{|x|^2}$

Question 43. Sketch each of the following:

a) $y = |(x-2)(x+5)|$
b) $y = |\ln x|$
c) $y = |\sin x|$ for $-\pi \leq x \leq 2\pi$
d) $y = |\cos x|$ for $-\pi \leq x \leq 2\pi$
e) $y = |x-3| + |x+4|$
f) $y = |x-2| + |2x+1|$
g) $y = |x+5| - |2x+7|$

Question 44. Sketch the following for $-360° \leq x \leq 360°$:

a) $y = \sin|x|$

b) $y = \tan|x|$

Question 45. It is given that $f(x) = x^3 + 6x^2 + 3x - 10$. Sketch on separate axes:

a) $y = f(x)$

b) $y = |f(x)|$

c) $y = f(|x|)$

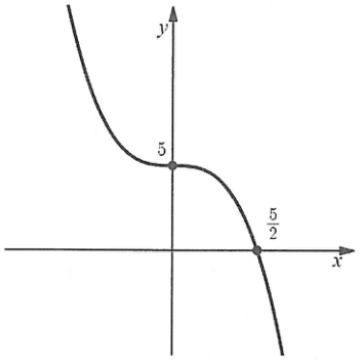
Question 46. It is given that $f(x) = \cos 2x$ for $-\pi \leq x \leq \pi$. Sketch on separate axes:

a) $y = f(x)$
b) $y = |f(x)|$
c) $y = f(|x|)$

Question 47. It is given that $f(x) = \tan 3x$ for $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$. Sketch on separate axes:

a) $y = f(x)$
b) $y = |f(x)|$
c) $y = f(|x|)$

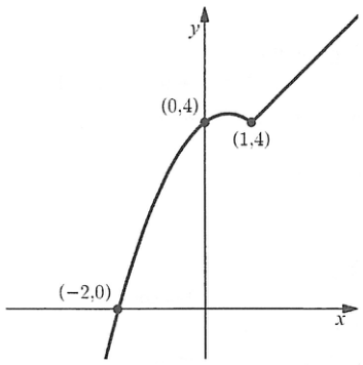
Question 48. The diagram below shows part of the curve with equation $y = f(x)$



The curve passes through the points $(0, 5)$ and $(\frac{5}{2}, 0)$. Sketch on separate axes:

a) $y = |f(x)|$
b) $y = f(|x|)$
c) $y = 2f(x)$

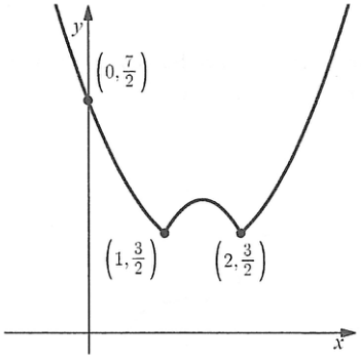
Question 49. The diagram below shows part of the curve with equation $y = f(x)$



The curve passes through the points $(-2, 0)$, $(0, 4)$ and $(1, 4)$. Sketch on separate axes:

a) $y = |f(x)|$
b) $y = f(|x|)$
c) $y = f(2x)$

Question 50. The diagram below shows part of the curve with equation $y = f(x)$



The curve passes through the points $(0, \frac{7}{2})$, $(1, \frac{3}{2})$ and $(2, \frac{3}{2})$. Sketch on separate axes:

a) $y = f(|x|)$
b) $y = f(\frac{1}{3}x)$
c) $y = 2f(2x)$

Question 51. It is given that $f(x) = x^2 + x - 6$

a) Sketch $y = |f(x)|$
b) Sketch $y = f(|x|)$
c) Find the set of values of x for which $f(|x|) = |f(x)|$

Question 52. It is given that $f(x) = \frac{a}{x}$ for $a > 0$

a) Sketch $y = |f(x)|$

b) Sketch $y = -f(|x|)$

c) Find the set of values of x for which $f(|x|) = |f(x)|$

Question 53. It is given that $f(x) = e^x - a$ for $a > 0$

a) Sketch $y = f(x)$

b) Sketch $y = |f(x)|$

c) Sketch $y = f(|x|)$

d) Find the set of values of x for which $f(|x|) = |f(x)|$

Question 54. Find the set of values for which

a) $x^3 = |x|^3$

b) $x^3 - 4x = |x^3 - 4x|$

c) $\ln x = |\ln x|$

d) $e^{x-1} - 4 = |e^{x-1} - 4|$

Question 55.

a) Write $-(x+2)^2 - 1$ in the form $ax^2 + bx + c$ where a, b and c are integers to be found.

b) Sketch $y = -(x+2)^2 - 1$ and $y = -|x|^2 - 4|x| - 5$ on the same axes

c) Hence find the set of value(s) of x such that $-(x+2)^2 - 1 = -|x|^2 - 4|x| - 5$

Question 56. Solve $x^2 - 4x + 5 = -|x|^2 - 4|x| + 6$.